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Comparative study on the pyrolysis behaviors of rice straw under different washing pretreatments of water, acid solution, and aqueous phase bio-oil by using TG-FTIR and Py-GC/MS

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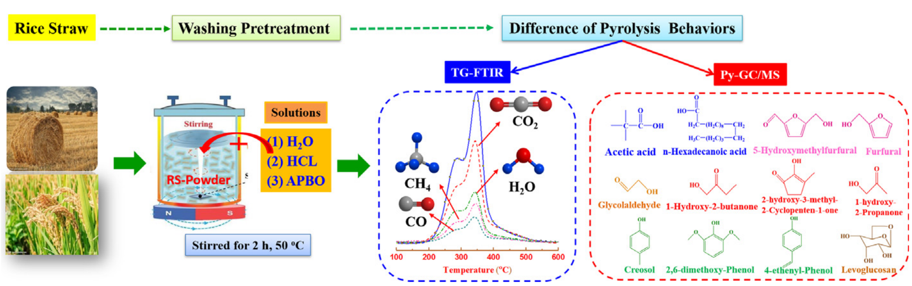
GRAPHICAL ABSTRACT



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ABSTRACT

Washing pretreatment is a promising method for upgrading biomass by removing alkali metal and alkaline earth metals (AAEMs). In this study, three types of solution with different pretreatment severities (i.e., water (H_2O), dilute hydrochloric acid (HCl) solution, and aqueous phase bio-oil (APBO)) were used to study the effects of the washing pretreatment on the pyrolysis behaviors and kinetics of rice straw. The pyrolysis experiments of raw and pretreated rice straw (Raw-RS, H_2O-RS, HCl-RS, and APBO-RS) were carried out using TG-FTIR and Py-GC/MS. Results show that among the three types of washing pretreatment methods, the APBO washing pretreatment has the highest removal efficiency of AAEMs: 99.7% of K, 91.7% of Na, 96.6% of Mg, and 95.2% of Ca. According to TG-FTIR results, only one sharp mass loss peak was present in the DTG curves of Raw-RS and H_2O-RS, whereas two mass loss peaks (one sharp peak and one shoulder peak) were observed in HCl-RS and APBO-RS. The evolution pattern of the volatile compounds (H_2O, CH_4, CO, CO_2, and C=O stretching) was thoroughly investigated, and the absorbance intensity of these components increased with the washing pretreatment of APBO and HCl. The APBO washing pretreatment had the highest activation energy in the conversion rate of 0.2–0.75, followed by HCl and H_2O washing pretreatment. According to the Py-GC/MS results, the washing pretreatment reduced the relative contents of acids, ketones, furans, and phenols in bio-oil, whereas removing of AAEMs increased the relative contents of anhydrosugars (mainly levoglucosan). Among the three types of washing pretreatments, the APBO washing pretreatment had the most remarkable influence on the property of bio-oil,

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and this indicates that the APBO washing pretreatment is a promising method for upgrading biomass and its pyrolytic product.

1. Introduction

As an important liquid product of biomass pyrolysis, bio-oil upgrading has obtained increased attention [1,2]. The ash in biomass contains a great variety of alkali and alkaline earth metals (AAEMs), such as Na, K, Mg, and Ca. The AAEMs act as catalysts during the biomass fast pyrolysis process, which promotes the generation of H_2O and acetic acid and has an adverse effect on the quality of bio-oil [3–5]. Therefore, it is essential to remove ash before the pyrolysis process to upgrade raw biomass and the subsequent pyrolysis products.

Washing pretreatment, including washing with water washing and acid, is the most commonly used method for removing ash in biomass. A dilute acid washing method (using inorganic and organic acids) was reported to be more effective than the water washing method at removing ash, and there was some accompanying damage to the chemical structure of biomass [6]. Carpenter et al. reported that even a minor amount of AAEMs in biomass remarkably alters the pyrolysis behaviors [7]. The acid washing pretreatment effectively increases the weight loss rate and decreases the yield of the solid product of biomass using TGA analysis. In terms of pyrolysis kinetics, the activation energy of rice husks slightly increased from 75.68 kJ/mol to 79.29 kJ/mol after an acid washing pretreatment using HCl [8]. A similar result was also found for three energy crops: switchgrass leaves, giant reeds, and cardoon leaves [9]. However, the release characteristics of volatiles during biomass pyrolysis and how activation energy varies with respect to the conversion rate have not yet been thoroughly investigated.

Acetic acid is a typical organic acid and has been proved to effective at removing AAEMs from biomass [10,11]. Interestingly, acetic acid is a typical component in bio-oil [12]. After a certain length of storage time, bio-oil usually separates into two layers. Specifically, the top layer is an aqueous phase bio-oil (APBO), and the bottom layer is an oil phase bio-oil (OPBO) [12,13]. Most of the hydrophilic acid components are contained in the APBO along with a small amount of phenols. Therefore, APBO might be a feasible substitute for commercial acids used in biomass that is pretreated with acid washing [8,14–18]. Karnowoet et al. found that removal of AAEMs, especially potassium (K), is a promising method for upgrading the feedstock of rice husks [17]. Zhang et al. found that the AAEMs in rice husks are effectively removed by washing with light liquid bio-oil [14]. Oudenhoven et al. [16] reported that washing biomass with wood-derived acid is beneficial for the selective production of levoglucosan. In addition, Chen et al. reported that a combined pretreatment of torrefaction and washing with APBO remarkably reduces the amount of water and acid components in bio-oil but increases the relative content of phenols [18]. These previous studies achieved remarkable advances in understanding the effects that washing pretreatments have on improving biomass and subsequent pyrolysis products. However, a comparative study on the effects of various washing pretreatments with different severities (water washing, acid washing, and APBO washing) on the thermal degradation behaviors and kinetics parameters of biomass using Py-GC/MS and TG-FTIR has not yet been reported.

Rice straw is a massive agricultural residue in the world, and it is a promising biomass for conversion into bio-fuels. In this study, three types of solution (water, HCl solution, and APBO) were used in the washing pretreatment of rice straw. The purpose of this study was to use TG-FTIR and Py-GC/MS to conduct a systematic comparison of the thermal degradation behaviors and kinetics parameters of washing-pretreated rice straw.

2. Materials and methods

2.1. Materials

In this study, rice straw was obtained from Anhui Province and was used as raw material for the washing pretreatment and pyrolysis experiment. The rice straw was milled into powder samples, and the powder was screened using 40–60 mesh sieves. The powder was then dried at 105 °C for 12 h before the washing experiment.

2.2. Washing experiments

Three washing solutions (water, dilute hydrochloric acid solution (pH = 2.9), and APBO) were used in the biomass washing pretreatment experiment. As reported in a previous study, APBO was separated from the bio-oil produced from rice straw pyrolysis at 550 °C in a lab-scale reactor [18]. The top layer of the bio-oil was APBO and had a pH value of 2.9. In each biomass washing experiment, about 3 g of rice straw was steeped in 100 mL of washing solution, and then the mixed solution was stirred at 50 °C for 2 h. Finally, to obtain the washed biomass, the solution was filtered and washed with distilled water until the pH of the solution became neutral. The dried rice straw was labeled as Raw-RS. The samples that were washed with the different solutions were labeled as H_2O-RS, HCl-RS, and APBO-RS.

The amounts of ash and AAEMs before and after the washing pretreatment are presented in Table 1. After the washing pretreatment, the amount of ash in the washed sample was remarkably decreased, and this was especially true for the APBO-RS sample. After the water washing pretreatment, most of the K in the biomass was removed. However, most of the other metallic species (Ca, Na, and Mg) remained in the biomass. In contrast, the four types of AAEMs were completely removed with both the HCl and APBO washing pretreatments. Among the three types of washing pretreatments, the APBO washing pretreatment had the highest removal efficiency of AAEMs; specifically, it removed 99.7% of K, 91.7% of Na, 96.6% of Mg, and 95.2% of Ca.

2.3. Pyrolysis experiments

2.3.1. TG-FTIR experiments

Weight loss characteristics and identification of volatile components of the washing-pretreated rice straw were determined using TG-FTIR

Table 1
Alkali and alkaline earth metallic species (AAEMs) content and demineralization efficiency of the samples.

Sample	AAEMs content (μg/g)				Demineralization efficiency (%)			
	K	Ca	Na	Mg	K	Ca	Na	Mg
Raw-RS	19175 ± 389	3521 ± 176	1362 ± 83	2107 ± 92	0	0	0	0
H_2O-RS	3283 ± 152	3108 ± 179	376 ± 75	1625 ± 51	82.88	11.73	72.39	22.88
HCl-RS	89 ± 11	218 ± 17	125 ± 12	58 ± 9	99.54	93.81	90.82	97.25
APBO-RS	58 ± 5	169 ± 19	113 ± 11	71 ± 8	99.70	95.20	91.70	96.63

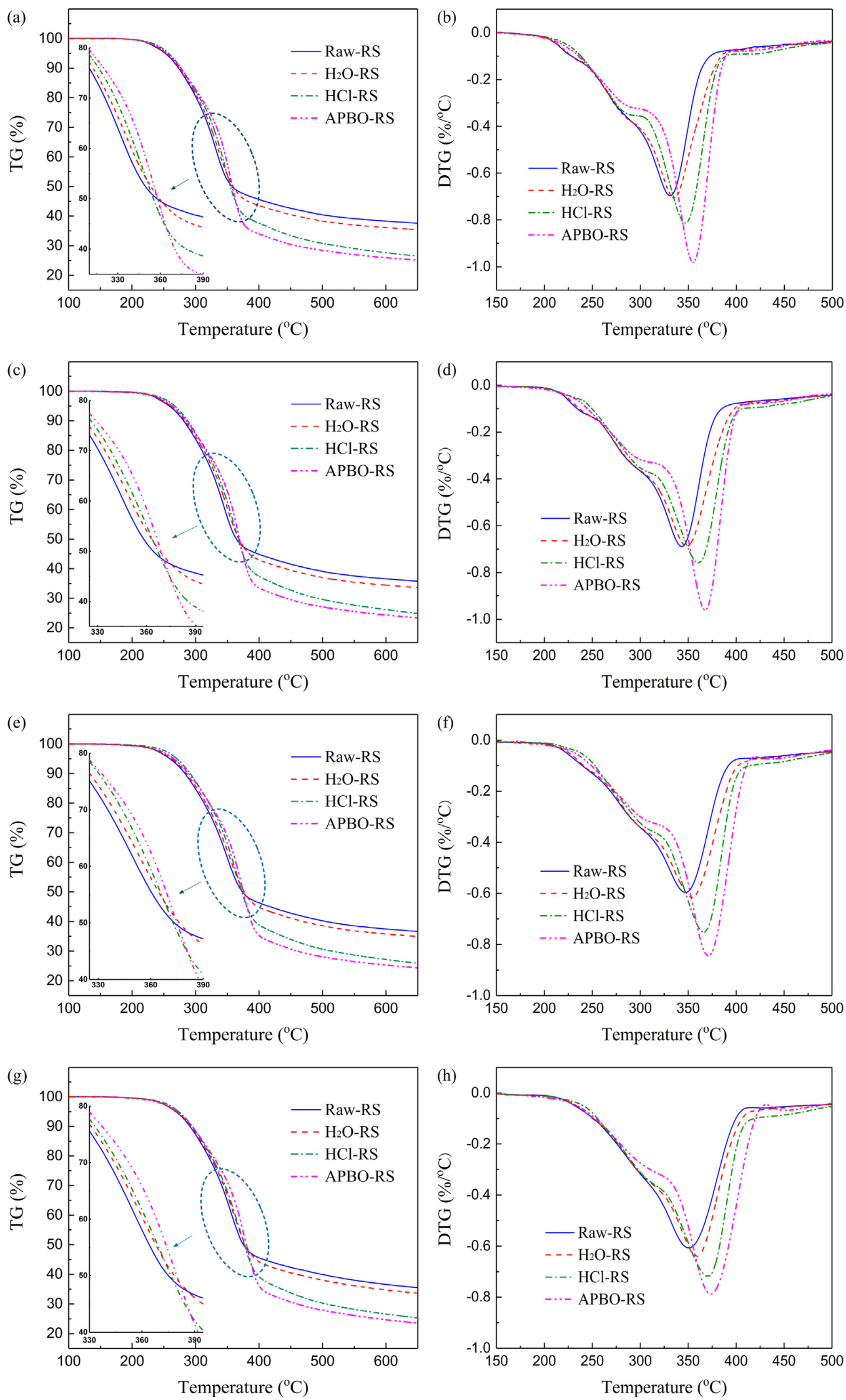

Fig. 1. TG and DTG curves of Raw-RS, H_2O-RS, HCl-RS, and APBO-RS at different heating rates: (a) and (b) at 10 °C/min, (c) and (d) at 20 °C/min, (e) and (f) at 30 °C/min, (g) and (h) at 40 °C/min.

(TGA Q500, TA Instruments, USA coupled with Nicolet 6700, Thermo Scientific, USA). The flow rate of high-quality nitrogen (carrier gas) was 70 mL/min. About 20 mg of rice straw was used in each experiment, and the heating program was set to increase from 50 to 650 °C at different heating rates (10, 20, 30, and 40 °C/min). Finally, thermogravimetric and differential thermogravimetric curves and three-dimensional (3D) FTIR spectra were recorded.

2.3.2. Pyrolysis kinetics analysis

Pyrolysis kinetic parameters of raw and pretreated rice straw were determined by the distributed activation energy model (shorten in DAEM), which was widely used for biomass kinetics analysis [19,20]. The model assumes that the thermal degradation of biomass is composed of an infinite number of first-order parallel reactions. Then the activation energy estimated from the TG data, is a continuous distribution function depending on the conversion rate (α) or temperature (T). The model is represented as Eq. (1):

$$1 - V/V^* = \int_0^\infty \exp\left(-\frac{k}{\beta}\int_0^T e^{-E/RT}dT\right)f(E)dE \tag{1}$$

where V is the volatiles at the pyrolysis temperature of T, V^* is the effective content of volatiles, $f(E)$ is the distribution curve of activation energy. Other symbols of E, k, and β represent the activation energy, the frequency factor, the universal gas constant, respectively. In this study, the conversion rate (α) is defined as V/V^*, and could be calculated by the Eq. (2):

$$a = \frac{m_0 - m_t}{m_0 - m_f} \tag{2}$$

where m_0, m_t, and m_f are the initial mass of samples, the actual mass of samples at the pyrolysis time of t, and the final mass of samples, respectively.

After a series of mathematical simplification and approximation, the DAEM model can be simplified to Eq. (3):

$$\ln\left(\frac{\beta}{T^2}\right) = \ln\left(\frac{kR}{E}\right) + 0.6073 - \frac{E}{R}\frac{1}{T} \tag{3}$$

The detailed mathematical process was presented by Miura and Maki [20]. At a specific conversion rate, the Arrhenius plots of $\ln(\beta/T^2)$ vs $1/T$ will be a straight line, from which E and k values will be obtained from the slope ($-E/R$) and the intercept ($\ln(kR/E) + 0.6073$). Generally, multi-heating rates are employed for a more accuracy value of activation energy.

2.3.3. Py-GC/MS experiments

A pyrolysis reactor coupled with a gas chromatography/mass spectrometer (denoted as Py-GC/MS) was used to identify the components of bio-oil obtained from pyrolysis of washing-pretreated rice straw. The pyrolysis reactor is a CDS 5250 (Chemical Data Systems, USA), and the GC/MS instrument is a Trace DSQ II (Thermo Scientific, USA). The raw and pretreated rice straw was pyrolyzed in 550 °C for 10 s at a heating rate of 10 °C/ms. More detailed information regarding the instrument and pyrolysis process can be found in a previously reported study [21].

3. Results and discussion

3.1. Thermogravimetric analysis

Fig. 1 shows TG-DTG curves of Raw-RS and the washing-pretreated samples (HCl-RS, H_2O-RS, and APBO-RS) at four heating rates. The thermal degradation characteristics of the four samples are listed in Table 2. The thermal degradation of each of the four samples was composed of three stages: a moisture evaporation stage (50–150 °C), a fast devolatilization stage (150–450 °C), and a carbonization stage (450–650 °C). A slight mass loss occurred in the first stage. Then, thermal degradation of cellulose, hemicellulose, and lignin of biomass began, and this resulted in a large percentage of mass loss. Chen et al. have conducted research on the pyrolysis mechanism of biomass (bamboo), and found that several weakly linked branched function groups easily fell off by a series of dehydration and decarboxylation reactions at 250–300 °C, resulting in the formation of CO_2, acetic acids, and H_2O [22]. Similar results were found for hemicellulose, cellulose, lignin, and rice husk [23,24]. At higher pyrolysis temperature of 300–350 °C, some phenolic components were produced by the breakage of ether bridges in the lignin. Finally, the mass loss rate gradually decreased, and the TG-DTG curves became almost flat in the carbonization stage.

For the three washing-pretreated samples, remarkable differences are observed in the TG-DTG curve shapes, shoulder peak, maximum mass loss rate, and final residue mass. Compared with the Raw-RS, the mass loss in the second stage had a wider temperature range for the H_2O-RS, HCl-RS, and APBO-RS. In addition, the maximum loss rate of H_2O-RS (0.62–0.71%/°C), HCl-RS (0.72–0.81%/°C), and APBO-RS (0.79–0.98%/°C) was higher than that of Raw-RS (0.61–0.70%/°C). APBO-RS had the highest weight loss rate (0.98%/°C), and this might be because of the maximum removal efficiency of AAEMs in the APBO washing pretreatment. Jiang et al. found that AAEMs that were present in the pore structure were removed via a leaching treatment with deionized water, acetic acid, hydrochloric acid, nitric acid, and orthophosphoric acid and that the average pore diameter of rice straw then increased [25]. This result indicates that acid washing pretreatments are beneficial for releasing volatile compounds in biomass, and thus, the residual solid mass reduced.

As seen in the DTG curves of Fig. 1, there was only one sharp peak in the fast devolatilization stage of Raw-RS and H_2O-RS. However, there are two mass loss peaks (one shoulder peak and one sharp peak) in the DTG curves of HCl-RS and APBO-RS. The shoulder peak is attributed to the pyrolysis of hemicellulose, whereas the sharp peak is attributed to the pyrolysis of cellulose [26]. The APBO and HCl washing pretreatments caused these two pyrolysis peaks to be more distinct. Meanwhile, the maximum mass loss rate increased with the removal of AAEMs, and the corresponding peak temperature shifted to high temperature. The temperatures that correspond to the sharp peak in the DTC curves for H_2O-RS (335.30–358.95 °C), HCl-RS (346.74–370.28 °C), and APBO-RS (354.96–373.43 °C) were 5 to 23 °C higher than that in the DTC curve

Table 2
Thermal degradation characteristic of Raw-RS, H_2O-RS, HCl-RS, and APBO-RS at different heating rates.

Samples	Heating rate (°C/min)	Shoulder peak	T_{max} of sharp peak (°C)[c]	DTG_{max} of sharp peak (%/°C)[d]
Raw-RS	10	/[a]	330.96	0.70
Raw-RS	20	/	343.19	0.69
Raw-RS	30	/	347.15	0.60
Raw-RS	40	/	350.52	0.61
H_2O-RS	10	/	335.30	0.71
H_2O-RS	20	/	348.53	0.69
H_2O-RS	30	/	353.96	0.62
H_2O-RS	40	/	358.95	0.64
HCl-RS	10	+[b]	346.74	0.81
HCl-RS	20	+	360.64	0.76
HCl-RS	30	+	366.25	0.75
HCl-RS	40	+	370.28	0.72
APBO-RS	10	+	354.96	0.98
APBO-RS	20	+	367.61	0.96
APBO-RS	30	+	370.89	0.85
APBO-RS	40	+	373.43	0.79

[a] Shoulder peak /: no obvious shoulder peak;
[b] Shoulder peak +: presenting of shoulder peak;
[c] T_{max}: the temperature of maximum loss rate;
[d] DTG_{max}: the mass loss rate of corresponding T_{max}.

for Raw-RS (330.96–350.52 °C) with the order of APBO-RS > HCl-RS > H_2O-RS > Raw-RS. These results are attributed to structural changes in the samples caused by washing pretreatments. Previous references have reported that the peak temperature of the DTG curve for cellulose pyrolysis shifts to high temperature because of the remove of AAEMs by dilute acids or hot water [27]. Similar results are found in the literature, indicating that the presence of AAEMs leads to a decrease in the temperature at the point of maximum mass loss [8,18,28]. Water washing pretreatment had no obvious influence on the organic composition and chemical structure of biomass. However, acid washing pretreatment resulted in a more amorphous and loose structure in hemicellulose and cellulose, and this caused thermal degradation of the biomass to occur more easily in the pyrolysis and combustion processes [29,30].

3.2. Three-dimensional FTIR spectral analysis

Three-dimensional FTIR spectra (3D FTIR) of Raw-RS and washing-pretreated samples (HCl-RS, H_2O-RS, and APBO-RS) are presented in Fig. 2(a–d). FTIR identifications of the pyrolysis peaks are shown in Fig. 2e. As seen in Fig. 2(a–d), there are three volatile stages when the pyrolysis time or temperature are increased, and the changes in the shapes of the volatiles were quite similar to the shapes of DTG curves. Some typical products were identified by the characteristic absorbance bands (Fig. 2e). The bands between 4000 and 3400 cm^{-1}, 3050–2650 cm^{-1}, and 2230–2000 cm^{-1} are ascribed to H_2O, CH_4, and CO, respectively. The characteristic absorbance bands at 2400–2240 cm^{-1} and 1890–1630 cm^{-1} are ascribed to CO_2 and C=O stretching, respectively. The C=O stretching vibration is highly related to organic compounds, such as ketones, acids, and aldehydes [31]. With an increase in the pyrolysis time or temperature, the amount of volatiles first increased, reached its maximum value, then gradually decreased.

The APBO and HCl washing pretreatments each had a remarkable influence on the FTIR spectra. Compared with the absorption intensity of C=O stretching in Raw-RS and H_2O-RS, that of C=O stretching in HCl-RS and APBO-RS increased; also, the shoulder peaks of HCl-RS and APBO-RS became more distinct. After the APBO and HCl washing pretreatments, AAEMs were greatly removed, and the amount of pyrolysis volatiles and their release rate increased. The increase in the amount of pyrolysis volatiles directly improved the yield of condensable pyrolysis products (bio-oil), but the generation of the biochar was inhibited because the removal of AAEMs reduced the secondary decomposition reaction of volatiles. The TG curves show that the mass of the final residues was ordered as Raw-RS > H_2O-RS > HCl-RS > APBO-RS.

3.3. Evolution of some typical gas components

3.3.1. Evolution of H_2O

AAEMs also has obvious effect on the gas products of biomass pyrolysis [32,33]. Evolutions of some typical gas components are shown in Fig. 2f. When the pyrolysis temperature was lower than 150 °C, no remarkable characteristic absorbance peaks of H_2O were observed, and this was because the rice straw powder had been dried at 105 °C overnight before the pyrolysis experiment. Generally, the characteristic absorbance peak of H_2O was mainly appeared between 200 and 400 °C and was accompanied by two striking peaks. Wang et al. claimed that the formation of H_2O during biomass pyrolysis was mainly caused by breakage of hydroxyl (−OH) groups that were in hemicellulose and cellulose [34]. Therefore, the release of H_2O in the first and second peaks might be attributed to the pyrolysis of cellulose and hemicellulose, respectively. This conclusion was also summarized by Ding et al. [35] and Ma et al. [36]. All of the washing pretreatments led

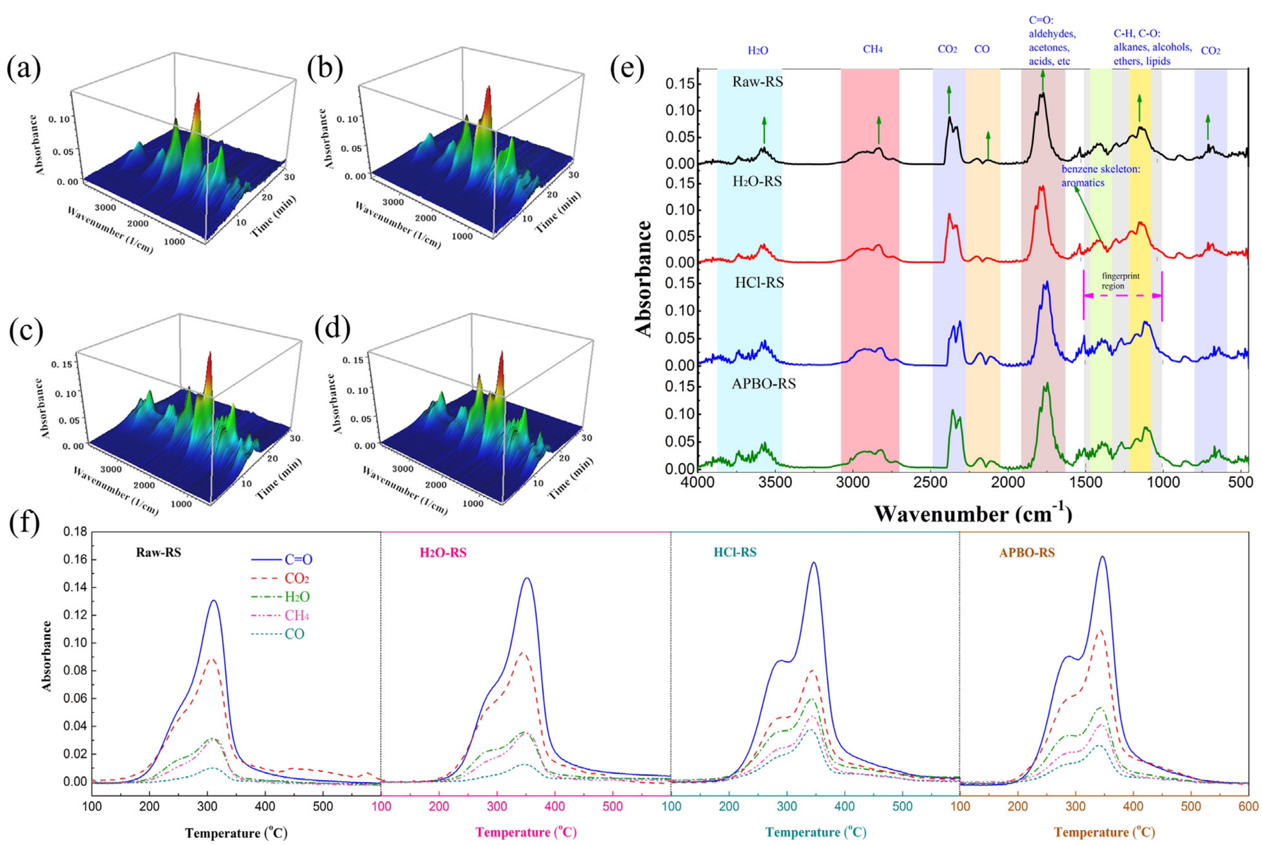

Fig. 2. 3D FTIR spectrograms of pyrolysis at the heating rate of 20 °C/min for the samples: Raw-RS (a), H_2O-RS (b), HCl-RS (c), and APBO-RS (d); FTIR identification of the pyrolysis peaks (e); and evolution of each detected gas component (f).

to an increase in the temperature that corresponded to the second peak of water and caused the separation of the two peaks to be more obvious. This occurred because the washing pretreatments (especially those with HCl and APBO) broke the crystalline region of cellulose and hemicellulose and caused the crystalline region to transform into an amorphous and loose structure [29,30].

3.3.2. Evolution of methane (CH_4)

Raw-RS and H_2O-RS did not show two distinct characteristic absorbance peaks of CH_4 in Fig. 2f. However, these peaks both appeared in the samples of HCl-RS and APBO-RS and mainly appeared in the temperature range of 200–400 °C. Wang et al., Ding et al., and Meng et al. [34,35,37] reported that methane was caused by the breakage of the methoxyl ($-OCH_3$), methylene ($-CH_2-$), and methyl ($-CH_3$) groups. The methoxyl group was mainly present in the phenylpropane basic unit of lignin, whereas the methylene and methyl groups were widely located on the side chain of the glucan units in hemicellulose, glycosyl units in hemicellulose, and phenylpropane units in lignin.

Among the three major components of biomass, lignin has been reported to have the widest thermal degradation temperature range (between 100 and 800 °C) [38,39]. Therefore, the first release peak of methane during pyrolysis of HCl-RS and APBO-RS might result from the breakage of the methoxyl ($-OCH_3$) group, which was weakly linked to the phenylpropane units of lignin. The ash removal efficiency of HCl and APBO was much higher than that of H_2O, and this resulted in a higher relative amount of lignin in samples of HCl-RS and APBO-RS compared to that in samples of H_2O-RS and Raw-RS. Therefore, the first evolution peak of CH_4 in each HCl-RS and APBO-RS was much more obvious than that in each H_2O-RS and Raw-RS. Thermal degradation of the three components of biomass intensified with an increase in the pyrolysis temperature and therefore accelerated the breakage of the methylene ($-CH_2-$) and methyl ($-CH_3$) groups and generated more methane. Thus, the second release peak of methane was most probably produced by the breakage of these two types of functional groups linked on the side chains of three major components in biomass.

3.3.3. Evolution of carbon dioxide (CO_2)

The evolution of carbon dioxide (CO_2) is also presented in Fig. 2f. There is one sharp evolution peak along with a shoulder peak. For Raw-RS and H_2O-RS, there were no obvious shoulder peaks, whereas for HCl-RS and APBO-RS, there were shoulder peaks observed in the range of 200–300 °C. Removal of AAEMs using dilute acids increased the temperature in correspondence with the maximum mass loss of cellulose pyrolysis, and this caused the DTG curve to shift to high temperature. For the first shoulder peak (which is at lower temperature), researches have indicated that CO_2 is mainly produced via two possible formation pathways [40,41]. The first formation pathway is the cracking of carbonyl ($-C{=}O-$) and carboxyl ($-COOH$) groups that are present in glucuronic acid of hemicellulose, and this cracking occurs via decarboxylation. Another formation pathway is the cracking of O-acetyl groups that are present in the xylan of hemicellulose via decarboxylation. The second sharp peak (which is at higher temperature) corresponds mainly to the thermal degradation of cellulose [42]. The ring-opening reaction of glucan units in cellulose is the dominant pathway for the formation of CO_2. The second decomposition reaction of the intermediate components (such as furans, acetones, and aldehydes) that are derived from the biomass primary pyrolysis also contribute some CO_2.

3.3.4. Evolution of carbon monoxide (CO)

Similar to the evolution of CO_2, in the evolution of CO for each HCl-RS and APBO-RS between 200 and 400 °C, there is one sharp peak along with a shoulder peak. In the lower pyrolysis temperature, the formation of CO can contribute to the cracking of ether bonds (β-O-4, 5-O-4, and α-O-4) in the lignin or the breakage of carboxyl groups present in the hemicellulose and cellulose. However, when the pyrolysis temperature was increased to over 400 °C, the formation of CO was mainly attributed to secondary decomposition reactions of aldehyde-containing intermediate components (such as aldehydes, furans, and furfurals) [35,37]. On one hand, HCl and APBO washing pretreatments led to an increase in the average pore diameter of biomass, and this was beneficial for the release of volatile compounds [25] and resulted in an increase in the intensity of CO. On the other hand, removal of AAEMs altered the thermal degradation pathway of rice straw, which resulted in an increase in the release of CO at lower pyrolysis temperature (200–400 °C).

3.3.5. Evolution of carbonyl (C=O)-containing components

The evolution of carbonyl (C=O)-containing gas components (acids, ketones, and aldehydes, etc.) is presented in Fig. 2f. The carbonyl-containing organics are primarily produced in ring-opening reactions of hemicellulose and cellulose. In addition, a small amount of (C=O)-containing organic compounds is a result of the breakage of the aliphatic lateral chain in lignin. Yang et al. [43] and Pasangulapati et al. [44] used TG-FTIR to investigate the evolution differences of carbonyl-containing organic compounds among the three major pseudo components (lignin, cellulose, and hemicellulose) in lignocellulosic biomass. For hemicellulose, the carbonyl-containing organic compounds appeared between 200 and 300 °C, whereas for cellulose, the carbonyl-containing organic compounds mainly appeared between 300 and 400 °C. In contrast to hemicellulose and cellulose, lignin only contributes a minor percent of carbonyl-containing organic compounds between 300 and 400 °C. Consequently, the first release peak of carbonyl-containing organic compounds is ascribed to the pyrolysis of hemicellulose for the HCl-RS and APBO-RS samples, whereas the second peak is a result of the pyrolysis of cellulose. As seen in Fig. 2f, the intensity of the shoulder peak at 280 °C was remarkably enhanced by the HCl and APBO washing pretreatments. This result indicates that removal of AAEMs in biomass is beneficial for the formation of carbonyl-containing organic compounds (such as acids, ketones, and aldehydes).

3.4. Pyrolysis kinetics analysis

Pyrolysis kinetics has been used widely to investigate the pyrolysis mechanism of lignocellulosic biomass in terms of kinetics parameters (the pre-exponential factor, activation energy, and mechanism function) [45]. From the TG/DTG curves, remarkable differences are found among the different washing pretreatment methods, which result in different activation energies. In this work, the activation energy (E) was determined by DAEM using TG data. The correlation between $\ln(\beta/T^2)$ and $1/T$ for the four samples at conversion rates of 0.15–0.9 are shown in Fig. S1 in the Supplementary Data. High correlation coefficients (R^2) were observed (values over 0.92), and these indicate a more accurate result for activation energy.

The activation energy of raw and washed rice straw is shown in Fig. 3. Except for H_2O-RS, the activation energy of rice straw samples increased with conversion rates in the range of 0.15–0.35 and then decreased with conversion rates in the range of 0.4–0.7; finally, the activation energy gradually increased with conversion rates in the range of 0.75–0.9. How the activation energy changed with an increase in conversion rate might be a result of the complex components (cellulose, hemicellulose, and lignin) because each component has its own devolatilization characteristics. Ma et al. [26] found that the thermal decomposition of hemicellulose occurs first at weakly linked sites, which thus results in very low activation energy; the activation energy then gradually increases because of random scission at the linear chain of cellulose at relatively high pyrolysis temperature. Thus, the activation energy gradually increased from 203 to 227 kJ/mol to 218–243 kJ/mol when the conversion rate increased from 0.15 to 0.35. According to the Broido–Shafizadeh cellulose pyrolysis model, which was proposed by Bradbury et al. [46], cellulose is initially transformed into active

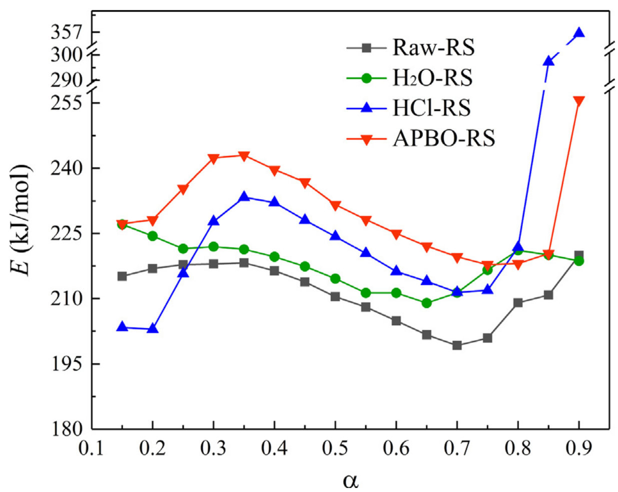

Fig. 3. Activation energy of Raw-RS, H_2O-RS, HCl-RS, and APBO-RS in the conversion rate range from 0.15 to 0.9.

cellulose with decreases in the degree of polymerization and molecular weight [47]. Then, lower energy is required for the pyrolysis of active cellulose with lower molecular weight, and this results in a decrease in the activation energy [36]. However, when the conversion rate was over 0.75, the activation energy increased remarkably because of the formation of biochar with dense carbon-rich structure that has low active of degradation [48].

In addition, Fig. 3 clearly shows that the activation energy increased after the washing pretreatment, and this is consistent with a previous study [8]. The activation energies of HCl-RS and APBO-RS are generally higher than those of H_2O-RS and Raw-RS over most of the range of conversion rates. Gao et al. reported that the presence of minerals leads to a decrease in the activation energy [28]. Therefore, in this study, most of the AAEMs in biomass were removed via the washing pretreatments of HCl and APBO, leading to an increase of pyrolysis activation energy of the HCl-RS and APBO-RS. In addition, an increase in activation energy requires higher reaction temperature, and this might better explain the slight shift in the DTG curves of HCl-RS and APBO-RS to high temperature.

3.5. Py-GC/MS analysis

According to the characteristic functional groups, the bio-oil identified by Py-GC/MS can be generally divided into several groups, such as aldehydes, acids, furans, ketones, and phenols. The relative contents of these groups that are derived from pyrolysis of rice straw using different washing pretreatment methods are presented in Fig. 4. The detailed components of bio-oil are listed in Table S1 in the Supplementary Data. The total content of bio-oil components was less than 100% because several components that are present in minute contents were not calculated. As seen in Fig. 4, the relative contents of phenols, ketones, acids, sugars, and aldehydes were 17.97–28.23%, 12.16–18.71%, 7.12–15.25%, 1.78–30.35%, and 9.09–11.71%, respectively. However, the relative contents of furans, esters, and N-containing compounds were each less than 6%. Compared with Raw-RS, the HCl and APBO washing pretreatments obviously decreased the relative contents of acids, ketones, and phenols. However, the relative content of anhydrosugars increased remarkably.

Changes in the composition of bio-oil from pyrolysis of rice straw using different washing pretreatments resulted from the catalytic effect of AAEMs. Zhang et al. [49] reported that K highly affects bio-oil components during biomass fast pyrolysis. Metal ions, such as K and Ca, are favored in reaction with oxygenated groups, promoting the ring conformation reaction, which remarkably increases the reactivity of cracking reactions [50]. In addition, a complex substrate of alkali metal ion-water (H_2O_M+, M: alkali metal) is produced by the interaction of alkali metal ions and functional groups of biomass (such as aldehyde and hydroxyl groups) [51]. Thus, the presence of K promotes dehydration during pyrolysis of rice straw. Meanwhile, a series of other low molecular weight components, such as furfural and phenols, increases because depolymerization and fragmentation of biomass by K are promoted [51,52].

Similar reaction between sodium (Na) ions and functional groups of biomass can also occur. Zhou et al. [53] added NaCl to glucose-based carbohydrates during pyrolysis and found that it promotes the thermal degradation of carbohydrates to produce more products with low molecular weight. Wang et al. also reported that magnesium (Mg) and calcium (Ca) have remarkable effects on pyrolysis of cellulose [6]. The addition of $MgCl_2$ and $CaCl_2$ increases the yield of furan derivatives (hydroxymethylfurfural, 2-furaldehyde, and levoglucosan) during pyrolysis of cellulose [54]. Thus, the washing pretreatment used in this study to remove AAEMs had an adverse effect on the formation of some chemicals (furans, ketones, acids, and phenols) [55].

In contrast, the relative content of anhydrosugars (such as levoglucosan) improved remarkably. Patwardhan et al. reported that the amount of added AAEMs, even at a very low catalyst/biomass ratio, highly inhibits the formation of levoglucosan during the pyrolysis of cellulose [54]. This occurs because the presence of AAEMs (K and Na) reduces the cracking of chains *via* end-chain initiation and depropagation [51,56]. This conclusion has also reported by other researchers, indicating that the relative contents of bio-oil components are highly dependent on the AAEM content in biomass [57].

As seen Table 1, the washing pretreatment (especially that with

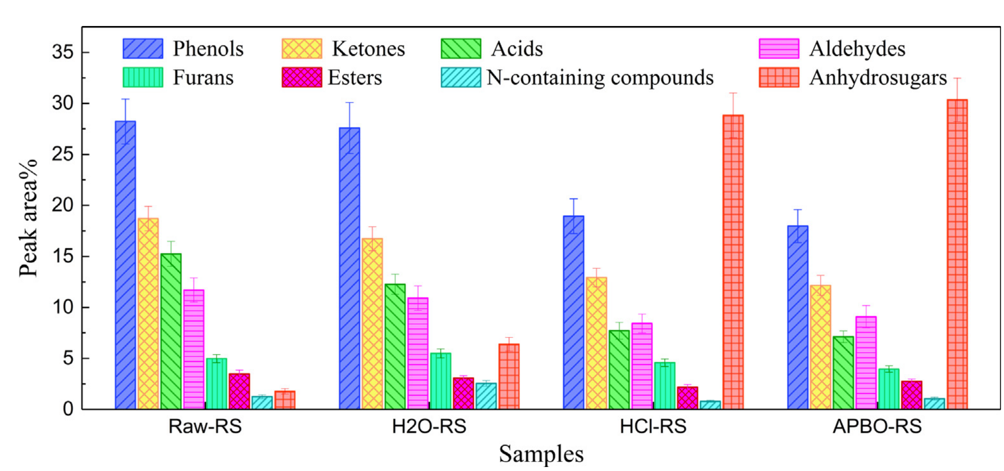

Fig. 4. Relative content of different groups of bio-oil generated from pyrolysis of Raw-RS, H_2O-RS, HCl-RS, and APBO-RS using Py-GC/MS.

APBO and HCl) can remove most of the K in biomass, and this results in an increased yield of levoglucosan. Pechaetal et al. [58] reported that levoglucosan is a high value-added chemical that can be used widely in the fields of polymer materials, medicine, and food. The relative content of levoglucosan from pyrolysis of APBO-RS was 30.35%, which is almost 17 times greater than that from pyrolysis of Raw-RS (1.78%). Thus, the relative content of acids was reduced after the washing pretreatment, whereas that of levoglucosan increased.

4. Conclusions

The effects of washing pretreatments using three types of solutions (water, dilute HCl, and APBO) on the pyrolysis behaviors of rice straw were investigated. Among the three types of washing pretreatments, the APBO washing pretreatment exhibited the highest removal efficiency of AAEMs (over 90%). Unlike the water washing pretreatment, the HCl and APBO washing pretreatments each had a remarkable influence on the pyrolysis behaviors of rice straw. In particular, these influences are observed in the TG/DTG curves, shoulder peaks, maximum loss rates, and 3D FTIR spectra. Most of the AAEMs were removed via the washing pretreatments of HCl and APBO, leading to an increase of activation energy in the HCl-RS and APBO-RS pyrolysis. APBO-RS had the highest weight loss rate and lowest final residue mass, indicating that the removal of AAEMs promoted the release of volatile compounds. The APBO washing pretreatment strongly promoted the formation of levoglucosan and inhibited the formation of acids, and these factors significantly improved the quality of bio-oil. Thus, the APBO washing pretreatment is a promising method for upgrading biomass and its pyrolytic liquid products.

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Declarations of interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2019.04.086.

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